

# A Transformer Based Approach for Activity Detection

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## ABSTRACT

Non-invasive physiological sensors allow for the collection of user-specific data in realistic environments. In this paper, using physiological data, we investigate the effectiveness of Convolutional Neural Network (CNN) based feature embeddings and Transformer architecture for the human activity recognition task. 1D-CNN representation is used for the heart rate, and 2D-CNN is used for short-term Fourier transformation of the accelerometer data. Post fusion, the feature is input into a transformer. The experiments are performed on the harAGE dataset. The findings indicate the discriminative ability of the feature-fusion on transformer-based architecture, and the method outperforms the harAGE baseline by an absolute 3.7%.

## CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing.**

## KEYWORDS

Human Activity Recognition, Transformers, CNN

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## 1 INTRODUCTION & BACKGROUND

The advancement of wearable devices equipped with robust sensors enables artificial intelligence-powered systems to track user activity. Identifying and forecasting human activities is critical for many environments such as surveillance, healthcare, and entertainment. Human Activity Recognition (HAR) seeks to detect and categorize human actions. HAR strategies are mainly classified into two types; vision-based and sensor-based approaches [4]. Vision-based methods are susceptible to occlusion, pose, and illumination changes [1, 9]. In addition, installing cameras in public places raises privacy concerns. However, sensor-based techniques seek to recognize human behavior using data gathered from wearable gadgets such as smartphones and smartwatches. These devices may accumulate

data from a broad user spectrum and utilize statistical learning techniques to categorize the signals into activity classes.

Commercial smartphone-based activity detection has substantially improved in recent years; smartphones now come equipped with in-built sensors like accelerometers and gyroscopes, making them suitable for identifying human activities. These sensors are non-invasive and wireless but suffer from orientation and placement-based concerns which can be challenging to accommodate in statistical models. On the other hand, smartwatch sensors provide superior performance in identifying a wider range of human actions due to their fixed positioning and orientation relative to the wrist.

The body undergoes physiological changes during physical activity, and wearable sensors can effectively record these changes over time. By carefully monitoring these physiological changes, one can estimate ongoing activities. An accelerometer is a commonly used sensor for tasks such as activity analysis in smart devices. It estimates acceleration forces to monitor movement. Data fetched from this sensor consists of 3-dimensional time series of data points, sampled at uniform intervals. Traditional use of manually generated attributes and heuristics from this data in developing statistical prediction models becomes challenging since there is no apparent analytical strategy to tie sensor data to the user's activity.

The authors in [5, 6] experimented with the outer product-based technique to unite the representations learned from these modalities after collecting physiological data (pedometer, accelerometer, and heart rate) from a commercial smartwatch. The multimodal models with CNN-based architectures that extracted deeply learned representations from input data from the accelerometer, heart rate, and pedometer modalities produced the most significant results. Their findings confirmed the task's viability for combining accelerometer, heart rate, and pedometer data with the suggested outer product-based strategy. The study detailed in [6] employs an end-to-end system and does not extract hand-coded features. It explores the six different strategies for fusing accelerometer data. The methods that include the accelerometer measurement's first and second-order derivatives produce better results in every fusion scenario, implying that the first two derivatives capture a suitable representation for identifying the activities under consideration. The authors also examined three different configurations, including CNN, RNN, and a combination of both. Their top-performing RNN+CNN network configuration noticed confusion with the sitting class; The windowed time samples related to this class were identified as belonging to the standing activity. The authors claim that a significant portion of the activity sequences were incorrectly categorized as walking, stair climbing, and hand washing. This outcome could be explained by the wrist stretching and shoulder roll motions, which the smartwatch's integrated accelerometer sensors would likely have trouble detecting.

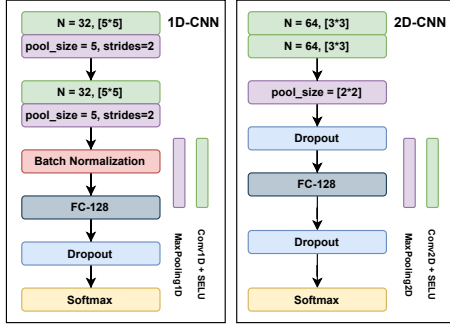
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**Figure 1: 1D CNN & 2D CNN architectures for extracting the embeddings from the heart rate and the accelerometer data.**

This study aims to examine the use of CNN and a transformer-based classification model to identify activities from a smartwatch’s accelerometer and heart rate data. In contrast to its predecessors, CNN automatically finds meaningful latent encodings without human oversight. The transformer’s attention-based encoder-decoder architecture efficiently classifies time series by producing position and context-aware representations from the input [8]. The primary contributions of this study for recognizing human activities from physiological data are as follows:

- We investigate the effectiveness of 1D-CNN when accelerometer and heart rate data are used independently.
- We investigate the utility of 2D-CNN on spectral images of accelerometer data.
- We demonstrate that the transformer model can successfully classify human activities; effective input embeddings for the transformer model are produced by combining feature embeddings derived from 1-D and 2-D CNNs on heart rate data and accelerometer spectral images, respectively.

## 2 DATASET

harAGE is a smartwatch-based multimodal data set for human activity recognition research. This dataset includes data samples collected from a commercial smartwatch sensor for eight different activities: *cycling, laying, running, sitting, stair climbing, standing, walking*, and *washing hands*. The dataset contains 20-second segments from embedded sensors such as accelerometers and heart rate monitors. The accelerometer data is sampled at 25 Hz, while the heart rate is sampled at 1 Hz. The dataset is divided into three participant-independent and gender-balanced train, development, and test partitions. The unweighted average recall (macro recall) is the metric for evaluating system performance. Readers interested in learning more about harAGE dataset can refer to [5, 10].

## 3 PROPOSED NETWORK ARCHITECTURES

We aim to investigate CNN and Transformers-based models for human activity recognition on commercial smartwatch sensor data. The activity recognition task consists of mapping the sensor’s data to predefined activity classes. We begin our investigation by examining 1-D CNN on raw accelerometer and heart rate sensor data. Later, we applied a short-time Fourier transformation (STFT) [2] to

extract the time-localized frequency information from accelerometer data. We fed frequency domain information into a simple yet efficient 2-D CNN to investigate the efficacy of frequency domain features in classifying activity classes. Additionally, we examined the performance of the standalone transformer, and a hybrid CNN-Transformer fusion model where latent encodings learned from CNNs are merged before being fed into the transformer model. We proceed into detail about the transformer model and CNN architecture in the following subsections.

### 3.1 1D-CNN

1D-CNNs are capable of extracting relevant embeddings, which encode the hidden sequential representations in time domain data [7]. An accelerometer sensor collects data as a three-dimensional time series, with each axis representing the effect of acceleration along the coordinate axis. To comprehend the hidden information from these time series, we employed a 1D-CNN model incorporating two stacked convolutional blocks. Each block comprises a convolutional layer with scaled exponential linear units (SELU) [3] activation. A max-pooling layer is positioned following the convolution layer to provide translation invariance to the internal representations. The batch normalization layer is applied after the second convolutional block to speed up training and minimize the internal covariate shift. To prevent overfitting, dropout regularization is applied following the fully connected layer. Figure 1 on the left illustrates the 1-D CNN model.

### 3.2 2D-CNN

Accelerometer signals are non-stationary. Converting an accelerometer sequence to the frequency domain produces a spectral image representing the change in frequency components over time; this representation reduces the effect of noise and enables better feature representation and modeling. We generate STFT images by passing accelerometer data through a short-term Fourier transformation. In order to extract hidden frequency-domain information, these spectral images are subsequently input into a 2-D CNN. The 2-D CNN architecture comprises two convolutional layers followed by a max pool operation. We apply scaled exponential linear units (SELU) [3] as an activation function. SELU has self-normalizing properties, which eliminate the chance of vanishing gradients. A dropout layer with a probability of 0.5 is placed after the fully connected layers to prevent overfitting. Figure 1 on the right illustrates the 2-D CNN model.

### 3.3 Transformer

Transformer neural networks are encoder-decoder architectures designed to solve sequence-to-sequence tasks. These architectures encapsulate the attention mechanism, which enables the discovery of relevant associations within the data and efficiently handles long-range dependencies. In this paper, we implement a transformer model inspired by the architecture mentioned in [11]. It comprises of an encoder and a classification head. Since we are working on a classification problem, the decoder part is entirely bypassed. Each encoder block comprises multi-head-self-attention layers, a feed-forward network, and includes dropout layers, layer normalization, and residual connections. A global average pooling layer and a

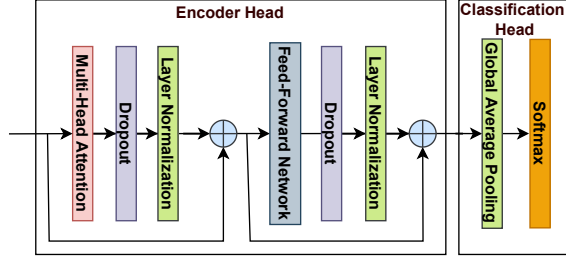


Figure 2: The block diagram of the encoder and classification heads of Transformer.

softmax layer construct the classification head. The global average pooling layer condenses encoder output into a single feature map, which softmax then transforms into a vector of probabilities. We empirically concluded that a stack of eight encoder blocks best suits our problem. Each encoder block has 16 multi-head-self-attention layers with the head size of 512 neurons and a feed-forward network with 128 neurons. The encoder and classification head dropout probabilities are set at 0.25 and 0.40, respectively. The Figure 2 illustrates the transformer encoder and classification head.

### 3.4 CNN-Transformer Fusion

Measuring bodily movements is insufficient for estimating a user's activity because it disregards the intensity of the activity and the user's effort. However, variation in heart rate measures incorporates vital information about body stress. We propose that estimation of activity class may benefit from a hybrid integration of heart rate and accelerometer sensor measurements. The proposed model aims to integrate the latent representations that modality-specific CNNs have learned. Figure 3 depicts the overall fusion architecture.

## 4 EXPERIMENTS

This section outlines the experiments conducted on the harAGE dataset. The experiments are coded in the TensorFlow framework. For all experiments, we are using constant epochs and batch sizes of 200 and 10, respectively. All models are tested on an unobserved development set after being trained on a predetermined train set.

### 4.1 1-D CNNs on Accelerometer & Heart Rate Data

For representing the heart rate and accelerometer data, respectively, we first evaluated a 1D-CNN. To also evaluate the discriminative ability of the features, we trained the shallow CNN with a categorical cross-entropy loss using the Nadam (adam with nesterov momentum) optimization.

### 4.2 2-D CNN on Accelerometer Data

We applied a 2-D CNN to extract pertinent features from the accelerometer based STFT images. To validate the discriminative ability of STFT images in their activity classes, we train the network on a pre-defined training set using adam optimizer on categorical cross-entropy loss. The learning rate is set to 0.0002.

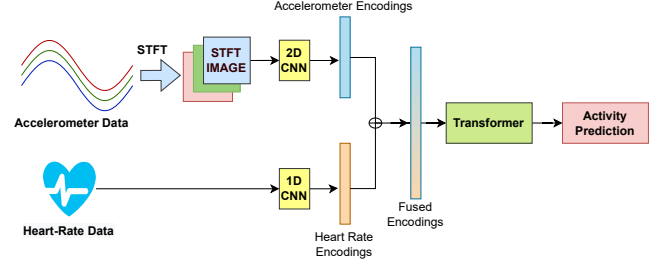


Figure 3: The block diagram of the proposed CNN-Transformer fusion model (Section 3.4).

### 4.3 Transformer on Accelerometer Data

Given the recent success of transformers in time series data, we directly apply transformer model to accelerometer data. This transformer is trained at a learning rate of 0.0002 using categorical cross-entropy loss and the Nadam optimizer. We classify the accelerometer data into its corresponding activity classes using softmax.

### 4.4 CNN-Transformer Fusion on Accelerometer & Heart Rate Data

We investigate the efficacy of fusing modalities that integrates the benefits of CNN and transformer. The proposed model's architecture is intended to handle both accelerometer signals and heart rate information. The objective is to merge the latent representations that modality-specific CNNs have learned before performing the final classification. We take heart rate data and pre-trained 1D-CNN to extract 128-dimensional latent embeddings. Additionally, on accelerometer STFT images, another set of 128-dimensional latent embeddings are retrieved using 2D-CNN. We concatenate these latent embeddings of accelerometer signals and heart rate data and use the transformer model to categorize them into the appropriate activity classes. The Nadam (adam with nesterov momentum) optimization approach is used to train this transformer model on categorical cross-entropy loss.

## 5 RESULTS & DISCUSSION

We validate the trained models using the harAGE dataset's unseen development set. The unweighted average recall (UAR) is the evaluation metric. The dataset organizer's baseline [10] achieved 72.2 % of UAR on the test set.

### 5.1 Overall Classification Results

In our experiments, we observe that all models outperformed chance UAR. The final column of the Table 1 displays the overall performance of the models on the development set.

**5.1.1 1-D CNN on Heart Rate Data:** The least effective model among all models is the 1D CNN when applied to heart rate data.

**5.1.2 1-D CNN on Accelerometer Data:** 1-D CNN performed significantly better on time-domain accelerometer data than on heart rate data; the results reveal a 12.85% gain in UAR score.

**Table 1: Comparison of various models on the harAGE dataset’s development set. The evaluation metric is the unweighted average recall (UAR).**

| Experiment(s)                | Cycling      | Laying       | Running       | Sitting      | Stairs Climbing | Standing     | Walking      | Washing Hands | UAR          |
|------------------------------|--------------|--------------|---------------|--------------|-----------------|--------------|--------------|---------------|--------------|
| 1D-CNN on Heart Rate         | 56.89        | 65.57        | 92.50         | 31.57        | 72.09           | 38.59        | 78.94        | 3.81          | 54.98        |
| 1D-CNN on Accelerometer      | 94.81        | 77.04        | 95.00         | 3.59         | 48.83           | 73.68        | 75.43        | 74.28         | 67.83        |
| 2D-CNN on Accelerometer      | <b>94.82</b> | 27.86        | 100.00        | 29.48        | 46.51           | 80.70        | <b>85.96</b> | 94.28         | 69.95        |
| Transformer on Accelerometer | 77.58        | 49.18        | 95.00         | <b>50.87</b> | 37.20           | 73.68        | 64.91        | 85.71         | 66.67        |
| CNN-Transformer Fusion       | 92.37        | <b>83.60</b> | <b>100.00</b> | 0            | <b>83.72</b>    | <b>85.96</b> | 84.21        | <b>97.14</b>  | <b>78.38</b> |

**5.1.3 2-D CNN on Accelerometer Data:** We observed a 2.12% improvement in UAR score when STFT images created from time-domain accelerometer data were processed with a 2D CNN.

**5.1.4 Transformer on Accelerometer Data:** Transformer model performed worse than 1D/2D-CNN on accelerometer data, but significantly better than 1D-CNN on heart rate data.

**5.1.5 CNN-Transformer Fusion on Accelerometer & Heart Rate Data:** When we applied the transformer model to the latent fused embeddings of the top-performing CNNs, we observed a significant increase in UAR, which empirically evaluates the effectiveness of this approach. We obtained a test UAR of 75.9, which is 3.7 points higher than the baseline results provided in the [10].

## 5.2 Classwise Classification Results

In this subsection, we take a closer look at the classwise examination of the classification results.

**5.2.1 1-D CNN on Heart Rate Data:** High-intensity activity classes like *running*, *stairs climbing*, and *walking* showed better performance than the others, with the *cycling* class exception. Heart rate sensor data appears to be highly discriminative for these high-intensity activities. The *washing hands* class performed the worst out of all classes.

**5.2.2 1-D CNN on Accelerometer Data:** Time domain accelerometer data seems to work best for *cycling* and *running* classes and reasonably well for *laying*, *standing*, *walking*, and *washing hands*. Out

of all the classes, the *sitting* and *Stairs Climbing* classes performed erroneously, with *sitting* having the lowest UAR score.

**5.2.3 2-D CNN on Accelerometer Data:** 2-D CNN trained on STFT images from accelerometer data detected *running* class perfectly. We also noticed significant performance improvement in the *standing*, *walking*, and *washing hands* classes. When comparing STFT images to raw time domain accelerometer data, there is a significant drop in *laying* class performance.

**5.2.4 Transformer on Accelerometer Data:** Compared to 1D-CNN on heart rate data, the transformer model on accelerometer data performed better in the *cycling*, *running*, *sitting*, *standing*, and *washing hand* classes. Except for the *sitting* class, the transformer model performed equivalent or lower for CNNs on accelerometer data. It is also worth noting that, compared to all other models we investigated, only the transformer model on accelerometer data performed well for the *sitting* class.

**5.2.5 CNN-Transformer Fusion on Accelerometer & Heart Rate Data:** CNN-Transformer fusion model is showing overall good results, but it is not at all able to identify *sitting* activity and confusing with *laying* and *standing*. Table 2 depicts the confusion matrix derived from CNN-Transformer fusion model on development data. In line with the observations in [6], there is a clear confusion between *standing*, *laying*, and *sitting* class. One possible explanation for this behavior is that body movement is kept to a bare minimum during all three activities, resulting in no distinguishable patterns that the model can learn from the accelerometer and heart rate encodings.

**Table 2: Confusion matrix for the CNN-Transformer fusion model (Section 5.2.5).**

| True Labels     | Predicted Labels |        |         |         |                 |          |         |               |
|-----------------|------------------|--------|---------|---------|-----------------|----------|---------|---------------|
|                 | Cycling          | Laying | Running | Sitting | Stairs Climbing | Standing | Walking | Washing Hands |
| Cycling         | 53               | 3      | 0       | 0       | 1               | 0        | 0       | 1             |
| Laying          | 0                | 51     | 0       | 0       | 0               | 9        | 0       | 1             |
| Running         | 0                | 0      | 40      | 0       | 0               | 0        | 0       | 0             |
| Sitting         | 0                | 42     | 0       | 0       | 0               | 13       | 0       | 2             |
| Stairs Climbing | 1                | 1      | 1       | 0       | 36              | 0        | 2       | 2             |
| Standing        | 0                | 7      | 0       | 0       | 0               | 49       | 1       | 0             |
| Walking         | 3                | 1      | 3       | 0       | 2               | 0        | 48      | 0             |
| Washing Hands   | 0                | 0      | 0       | 0       | 0               | 0        | 1       | 34            |

## 6 CONCLUSION

In this paper, we investigate the intricacy of human activity recognition using heart rate and accelerometer sensors, as well as the combination of both modalities. In comparison to heart rate data, the accelerometer data seems to be more appropriate. We propose a CNN-Transformer-based feature fusion technique for recognizing human activities. The proposed approach extracts relevant features from the accelerometer and heart-rate sensors using CNNs, and the transformer model receives input from these modality-specific CNNs. We observed that the CNN-Transformer fusion model performs better than the individual CNNs and transformers. Although the fusion network performs better overall, it is still inferior to standalone networks for *cycling*, *walking*, and *sitting* activities. Our findings encourage further research into incorporating feature-fusion architectures on physiological sensors for classifying human activities.

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